



Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Case Study: Bagging and the Spread Cross-Section

WEEK 1 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: a forest for a number

Morning: two hands-on sections

5.1 Regression trees and forests on the benchmark bonds. 5.2
The threshold rule, moved onto the training rows.

Afternoon: a case study

OLS, a depth-4 tree, and a 300-tree forest on the same bonds,
the same folds, the same metric.

Does an average of trees describe the spread cross-section better than Course 4's straight line?

The answer depends on which columns the models get, and on how the bonds are split.

Regression trees and forests

A leaf gives the mean log spread of its training bonds. The depth-4 tree asks score ≤ 9.5 first; its 16 leaves give 16 fitted spreads to 615 bonds. The first bond's leaf holds 56 of them.

0.8816

oob_score_ of a regressor: R-squared on log_oas

41.0 bp

MAE out of bag

16.2 bp

MAE on the bonds each tree was fitted on

40.6 bp

MAE from five folds

```
RandomForestRegressor(n_estimators=300, oob_score=True, random_state=SEED, n_jobs=1)
```

Day 2: oob_score_ of a classifier is accuracy. Same name, different measure: check it.

The threshold rule, on the training rows

1. Five folds

Each training row gets a probability from a model that never saw it.

2. Choose

The highest threshold with out-of-fold recall of at least 0.70.

3. Report

Fit on all training rows; flag the validation rows at that threshold.

Card file	Threshold	Out-of-fold recall	Validation recall	Validation precision
Logistic regression	0.2047	0.7002	0.6956	0.3319
Forest, 100 trees	0.1972	0.7002	0.6971	0.4128

Each model gets its own threshold. Validation recall lands near 0.70, not on it: an honest number.

Case study: the spread cross-section

Three models, Course 4's 615 training bonds, the same five folds, cross-validated MAE in bp. Without and with coupon; at random and by issuer.

OLS: Course 4's pipeline

Rating and sector as indicator columns; the numbers scaled.

Trees: the rating as its number

score from 1 (AAA) to 18 (CCC), sector as indicators, the numbers as they are.

Allowed names

fitted spread

the model's spread for these characteristics

residual

Never called

fair value or fair spread

cheap, rich, mispriced

a signal or a trade

Three models without coupon

40.0 bp

300-tree forest

41.7 bp

OLS, Course 4's pipeline

45.2 bp

one depth-4 tree

Mean of five fold MAEs in bp, random folds, on Course 4's training bonds

Read the fold sd first

Fold sd 11.6 bp for the forest, 10.7 bp for OLS: some folds hold harder bonds.

Then compare fold by fold

On the same bonds, the forest misses by less in 3 of 5 folds.

Averaging trees beats one tree by far, and the straight line by a little.

What the forest is given

42.3 bp

the forest on Course 4's one-hot columns (29 columns)

40.0 bp

the same forest on the rating's number (14 columns)

41.7 bp

OLS, for reference

A straight line gives each rating its own level. A tree spends one question per rating level when the rating comes as indicators; the rating's number lets it ask about many levels at once.

What a model is given matters as much as which model it is.

Adding coupon

MAE bp	Without coupon	With coupon	Gain from coupon
ols	41.7	37.2	4.5
tree_depth4	45.2	43.8	1.4
forest_300	40.0	38.3	1.7

The generator sets the coupon from the yield, and the yield holds the bond's own spread: coupon partly restates the target (Course 4 day 5).

With coupon, the straight line misses by less than the forest.

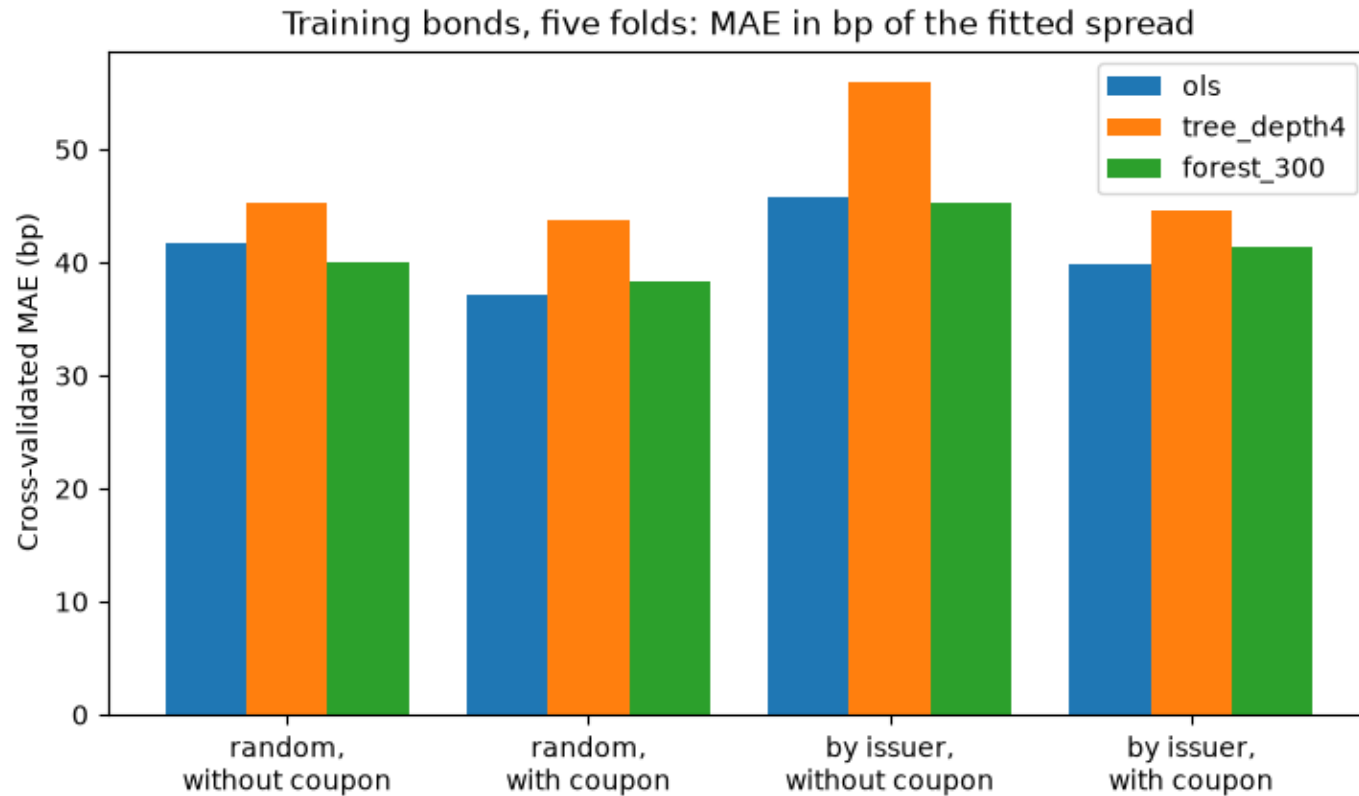
Splitting by issuer

Without coupon, MAE bp	Random folds	By issuer	Cost of a new issuer
ols	41.7	45.8	4.0
tree_depth4	45.2	56.0	10.7
forest_300	40.0	45.2	5.2

GroupKFold(5) with groups=ticker keeps each issuer's bonds in one fold. 6 warnings: ratings held by one issuer only are unknown to OLS's encoder when that issuer is scored.

At random: a bond of an issuer the model has seen. By issuer: an issuer it has not.

The comparison



Without coupon

The forest is lowest, at random and, only just, by issuer.

With coupon

OLS is lowest under both splits. The depth-4 tree is behind everywhere.

What the model shows, and where it fails

What it shows

Forest 40.0 bp, OLS 41.7 bp, one tree 45.2 bp without coupon; with coupon, the line is ahead.

New issuers

Every model misses by more when the issuer was never seen.

Wide spreads

Forest MAE 79.9 bp on high yield, 19.2 bp on investment grade.

The edges

Widest OAS 2,074 bp; the forest's widest fitted spread 915.0 bp, OLS 1025.9 bp.

The data

Synthetic bonds, one date. A fitted spread is never a fair value.

Excel side by side

Excel side by side (tested in Excel)

`=EXP(C2)`

`=AVERAGE(ABS(EXP(B2:B616) - EXP(C2:C616)))`

`=AVERAGE(F2:F6)`

`=STDEV.P(F2:F6)`

`=STDEV.S(F2:F6)`

`=1 - SUMXMY2(B2:B616, C2:C616) / DEVSQ(B2:B616)`

a fitted log spread in bp, filled down

MAE 39.9947 bp

mean of five folds 39.9947

fold sd 11.6052, as printed

12.9750: divides by 4, not printed

out-of-bag R-squared 0.8816

No regression tree, forest, or cross-validation in Excel. The scores are arithmetic, once the fitted values are pasted.

Save the week: git tag

1. Module

mlkit.py, ensemblekit.py and its tests

2. Random folds

results.csv: six rows, rows = cv

3. By issuer

six more rows, rows =
cv_by_issuer

4. Tag

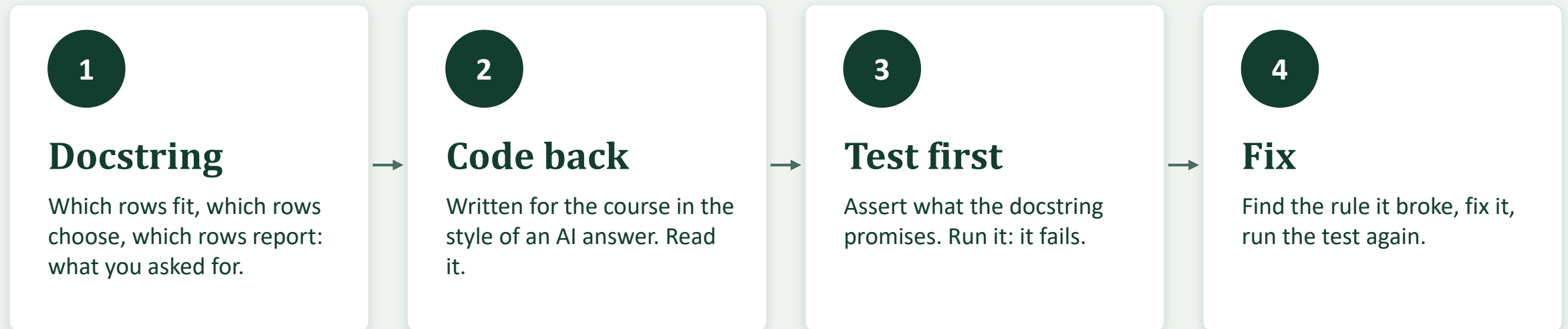
git tag week1

```
git diff week1~1 week1 -- results.csv
+ols,without coupon,cv_by_issuer,45.7623,0.8117
+tree_depth4,without coupon,cv_by_issuer,55.9545,0.7893
+forest_300,without coupon,cv_by_issuer,45.2195,0.8506
```

week1~1 is one commit before the tag: the diff shows the six rows by issuer. 21 tests pass at the end of today.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: the forest is cross-validated on all 821 bonds: Course 4's closed test bonds are scored, and the MAE (38.1 bp) looks fine.

Code that runs is not the same as code that is right.

What you can do now

- A regression tree's leaf is a mean; `oob_score_` is R-squared
- Thresholds from out-of-fold probabilities on training rows
- Read the fold sd before the gap; compare fold by fold
- Give a forest its inputs in a form a tree can use
- Split by issuer for the question about new issuers

Week 2, Day 1 Boosting begins: AdaBoost